



Therapeutic approaches employing natural compounds and derivatives for treating bipolar disorder: emphasis on experimental models of the manic phase

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Abstract

Bipolar disorder (BD) is a complex psychiatric disease characterized by mood swings that include episodes of mania and depression. Given its cyclical nature, BD is especially hard to model; however, the standard practice has been to mimic manic episodes in animal models. Despite scientific advances, the pathophysiology of BD is not fully understood, and treatment remains limited. In the last years, natural products have emerged as potential neuroprotective agents for the treatment of psychiatric diseases. Thus, the aim of this review was to explore the therapeutic potential of natural compounds and derivatives against BD, taking into account preclinical and clinical studies. Reliable articles indexed in databases such as PubMed, Web of Science and Science Direct were used. In clinical studies, treatment with herbal plants extracts, omega-3, inositol, n-acetylcysteine and vitamin D has been associated with a clinical improvement in symptoms of mania and depression in BD patients. In animal models, it has been shown that red fruits extracts, curcumin, quercetin, gallic acid, alpha-lipoic acid and carvone can modulate many neurochemical pathways involved in the pathophysiology of manic episodes. Thus, this review appointed the advances in the consumption of natural compounds and derivatives as an important therapeutic strategy to mitigate the symptoms of BD.

Keywords Mania · Bioactive compounds · Natural products · Neuroprotection · Bipolar disorder

Introduction

Bipolar disorder (BD) is a severe, chronic, and highly debilitating illness, which affects 1–4% of the world's population and is associated with significant morbidity and mortality risk (Price and Marzani-Nissen 2012; Saunders and Geddes 2016; Humpstom et al. 2021). BD is the sixth main cause of disability among physical and psychological disorders worldwide, leading to significant economic burden to patients and populations. It is characterized by recurrent pathological mood disturbances that range from extreme elation or mania/hypomania to severe depression (Young and Dulcis 2015; Carvalho et al. 2020). Considering that the biological basis of BD is unknown and its treatment have limited effectiveness and many side effects, the research for new therapeutic approaches for this psychiatric disease has been the focus of many studies in the last years.

Natural products, which derivatives from natural sources such as plants, animals and microorganisms have been intensively studied for their therapeutic potentials for many

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diseases (Nabavi et al. 2017; Chen et al. 2018; Sairazi and Sirajudeen 2020; Patil et al. 2020). Due the chemistry diversity, capacity to interacting concurrently with the multiple targets and few adverse effects, natural products and derivatives have drawn the attention of the researchers as a potential source to discovery and development to new drugs for both prevention or treatment to many pathological conditions, specially those diseases with a multifactorial etiology or variable response to treatment, such as BD. Thus, the aim of the present paper was to review the available literature, regarding the potential effects of natural compounds and derivatives in BD taking into account the results of preclinical and clinical studies.

The literature search was conducted using PubMed, Science direct, Google Scholar, and Web of Science as online databases until March 2021. In regard to studies about natural compounds and derivatives and BD, only papers written in English were considered with unlimited publication dates. The following combination of search terms were used in databases in english language: (bipolar disorder OR mania OR hyperlocomotion OR manic-like) AND (natural compounds OR plants OR fruit OR extracts OR vitamin OR fatty acid OR protein OR natural antioxidant OR flavonoids OR phenolic compounds OR derived from plants).

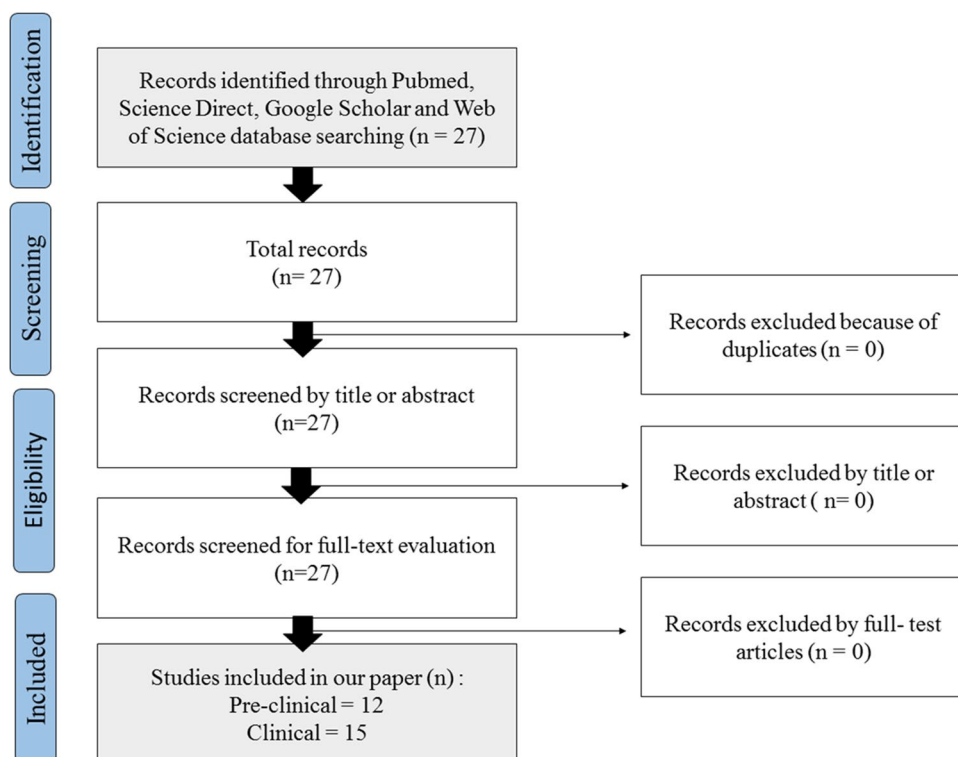
Two investigators independently screened the titles and abstracts of the articles to evaluate eligibility for inclusion until cos reached. Articles included should meet all the following criteria: (1) reviews and cell studies were excluded;

(2) original research with animal models and human was included; (3) patients included in the studies should have diagnoses for BD; (4) the intervention of interest was natural ou derived from natural compounds; (5) pre-clinical protocols must be validated through behavioral tests; (6) odds ratio (ORs) or relative risks (RRs) or hazard ratio (HRs) with 95% confidence intervals should be reported in the papers. Finally, 15 human and 12 animal models articles were included in the review. Flow diagram with the details of the study selection is shown in Fig. 1.

Bipolar disorder

Bipolar disorder (BD) is a mental disease characterized by recurrent pathological mood disturbances that range from extreme elation or mania/hypomania to severe depression (Carvalho et al. 2020). Manic episodes are associated with abnormally elevated mood, irritability, distractibility, psychomotor activation, and reduced need for sleep (Mcintyre et al. 2020). Hypomania is similar to the manic phase, but the symptoms are less severe than full mania. Depressive episodes are related to the presence of symptoms such as deep sadness, loss of energy, lack of interest in activities, fatigue, and suicidal thoughts, leading to significant suffering and/or impairment in social, professional, and other important facets of life (Mcintyre et al. 2020).

Fig. 1 Flowchart indicating the results of article for inclusion in this review



BD can be further subdivided into bipolar disorder type I, which is characterized by episodes resulting from mania and depression, with at least one manic episode throughout life, and bipolar disorder type II, which is characterized by alternations between episodes of hypomania and depression. In addition, cyclothymic disorder type, the mildest form of the disease, involves mood swings between hypomanic and depressive periods (Mcintyre et al. 2020). Males and females are equally affected by type I, whereas type II appears to be more common in women (Karanti et al. 2020). According to epidemiological data, BD starts on average between 15 and 44 years of age and has not consistently been associated with sociodemographic factors (Merikangas et al. 2011; Young and Dulcis 2015).

Although BD has been extensively studied, the causes and mechanisms involved in the pathogenesis of this disease are still unknown. Several hypotheses appear to be involved in the pathogenesis of BD, such as genetic and environmental factors, alterations in neurotransmitter signaling and neurotrophic factors, inflammation, and oxidative stress (Sigitova et al. 2017) (Fig. 2).

Epidemiological studies have demonstrated that affective disorders are heritable, and that first-degree relatives of affected individuals have a tenfold increased risk of developing BD compared to relatives of unaffected controls (Rowland and Marwaha 2018). In addition, environmental factors such as abuse, mental stress, a significant loss, or some other traumatic event may contribute to or trigger this disease (Aldinger and Schulze 2017).

Previous studies have demonstrated changes in dopaminergic, glutamatergic, noradrenergic, and serotonergic signaling in BD (Sigitova et al. 2017). In this context, an increase in dopamine levels has been observed in patients during manic episodes. In addition, an increase in the level of norepinephrine in manic individuals and reduced serotonin levels associated with suicidal aggressiveness and attempts have been reported (Ackenheil 2001). Significant enhancement in glutamate concentrations can occur in the prefrontal cortex of bipolar patients (Michael et al. 2003; Dager et al. 2004) suggesting that changes in glutamatergic neurotransmission are also involved in the etiology of this mood disorder.

Brain-derived neurotrophic factor (BDNF) is the most abundant neurotrophin in the brain of adult mammals and plays a central role in synaptic plasticity and neurogenesis (Song et al. 2017). This neurotrophic factor is found throughout the brain with higher abundance in the hippocampus and cerebral cortex, brain regions responsible for the control of mood, emotion, and cognition (Ernfors et al. 1990). Serum BDNF levels are reduced during manic and depressive episodes and recover after treatment for acute mania (Grande et al. 2012; Yu-Jie and Tiao-Lai 2019).

In recent years, studies have attempted to identify the mechanisms involved in the inflammatory processes involved in BD (Colpo et al. 2018; Teixeira et al. 2016). Data from the literature have been replicated and been shown to be consistent across studies, including the increase in the levels of pro-inflammatory cytokines in patients, particularly interleukin-6 (IL-6) and tumor necrosis factor-alpha (TNF- α), both in manic and depressive episodes (Muneer 2016; Fries et al. 2019). In addition, oxidative stress has also been implicated in the pathophysiology of BD. Oxidative stress is characterized by an imbalance between the production of reactive oxygen species and antioxidant defenses (Jones 2008; Andreatza et al. 2008). Patients with BD have significant alterations in antioxidant enzymes and lipid peroxidation (Andreatza et al. 2008). In fact, BD patients showed an enzymatic decrease in superoxide dismutase and catalase as well as in glutathione levels when compared to a control group, especially during acute disease episodes (Aydemir et al. 2014; Ghanizadeh and Berk 2013; Singh et al. 2019). Another study also reported significant changes in oxidative stress parameters in the blood of patients with BD according to the severity of the disease and not the number of manic episodes (Akarsu et al. 2018). Taken together, these evidences demonstrated that oxidative stress markers are increased in BD, indicating that alterations in redox status is associated with neurochemical changes observed in this psychiatric disorder (Mansur et al. 2016; Chowdhury et al. 2017; Jiménez-Fernández et al. 2021).

Mood stabilizers are the primary treatment for BD. Lithium has been the gold standard for the treatment of BD for

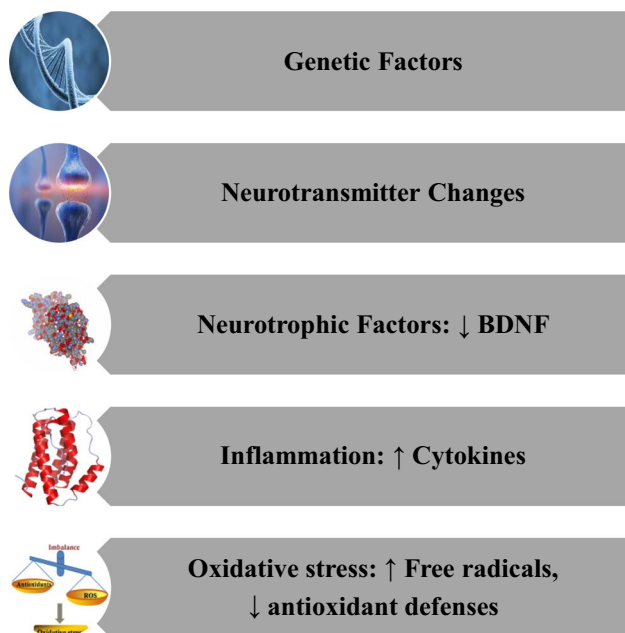


Fig. 2 Factors associated with the pathophysiology of bipolar disorder (BD)

several decades (Nolen 2015; Oruch et al. 2014). In addition to its mood-stabilizing properties, lithium exerts anti-suicidal, immunomodulatory and neuroprotective action (Rybakowski 2018). Lithium can be administered alone or in combination with other drugs, such as valproate and carbamazepine and lamotrigine (Chiu et al. 2013). At therapeutic doses, these medications generally require 1 to 4 weeks before their full effects are observed (Bauer and Mitchner 2004). Moreover, all medications used for treating BD have several adverse effects (Torpy et al. 2009), which contributes negatively to low patient adherence to treatment (Kasper 2014). Because BD is a chronic illness, continuous treatment is necessary to prevent relapse of manic or depressive symptoms, to improve overall health, and to maximize the quality of life.

In summary, although numerous mechanisms have been proposed to explain the pathophysiology of BD, the contribution of these mechanisms to specific symptomatology of this debilitating psychiatric ailment remains unknown. Additionally, several factors make the therapeutic management of BD complex, including the fluctuation in mood episodes and the effects of these episodes on the patient's well-being, treatment non-adherence, and comorbid psychiatric disorders. Thus, the research for new therapeutic approaches based in natural compounds has been the focus of preclinical and clinical studies in the last years. These studies can be a valuable starting point to discovery and development of more effective therapy to the management of BD.

Evaluation of natural compounds and derivatives in preclinical studies

Experimental models of mania

A manic episode is defined as a distinct period during which patients experience an abnormally and persistently raised, expansive, or irritable mood. These changes must be present for a period of one week or less. During this period, other symptoms are also observed, such as inflated self-esteem, psychomotor agitation, distractibility, involvement in activities with potential for painful consequences, and less need for sleep. The mood disorder is severe enough to cause impairments in occupational or social functioning, or to require hospitalization (Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition—DSM-5). Although manic episodes and hypomanic episodes have many similar symptoms, mood disturbances in hypomanic episodes are not sufficiently severe to cause pronounced impairment in social or occupational functioning. Mania and hypomania are two distinct types of episodes, but they have the same symptoms; however, mania is more severe than hypomania (Mcintyre et al. 2020). Animal models are essential tools in

preclinical research since they promote the advancement of knowledge of the neurobiology of several diseases. However, establishing an adequate model for BD is difficult because of its cyclical nature (Beyer and Freund 2017). It is also worth noting that some symptoms used to diagnose psychiatric illnesses in humans are not possible to assess in animals (Nestler and Hyman 2010). Because manic episodes are decisive for the diagnosis of BD, animal models that mimic this complex behavioral state are widely used to study this disease (Sharma et al. 2016; Beyer and Freund 2017).

It should be noted that several criteria exist for the evaluation of animal models for neuropsychiatric disorders. The animal models must meet these criteria, which demonstrate the relevance of translating experimental findings, in order to be validated. Briefly, animal models must have similarities with the symptoms of the disease (face validity), resemble pathophysiological aspects of the disease (construct validity), and respond to existing clinically successful treatments (predictive validity) (Willner 1984; Nestler and Hyman 2010; Valvassori et al. 2013).

Table 1 describes animal models of mania. It is worth mentioning that this table does not describe all existing mania models, but only those that satisfy the three criteria necessary to be validated (Sharma et al. 2016). Manic-like behavior in animals can be induced by different stimuli and divided into three types: pharmacological, environmental, and genetic. Table 1 displays the neurochemical mechanism that mimics the manic episode of each animal model, the behaviors used to evaluate that episode, and the referenced study. In addition, it should be noted that different tests were performed to assess the behavior of the animals. However, all the authors assessed locomotor behavior, making it possible to examine hyperactivity, one of the symptoms of this disease that can be reproduced in animal models (Ghedim et al. 2012).

In fact, the development of preclinical research is essential for knowledge about the pathophysiology of diseases and for the discovery of new pharmacological therapy. Considering that BD is a devastating mental illness, the pathophysiology underlying mania remains unknown and treatment is still considered limited, it is clear that this disease deserves special attention. An limitation related to the use of animal models for BD is the cyclical nature of this illness. So, the standard practice is to mimic the episode of mania and depression separately. There are several good approaches to model in rodents; however, most of these focus on either mania or depressive-like behavior. In an attempt to better understand the neurobiology of this disease in humans, an ideal model should demonstrate the episode of mania and depression in the same animal in a cycling manner. In addition, a wide variety of behavioral tests are also required for broader investigation, such as tests that investigate emotional states (Freund and Juckel 2019).

Table 1 Animal models of mania: neurochemical mechanism and behavioral tests

Animal model	Neurochemical mechanism	Behavioral test	References
Pharmacological Psychostimulant	Increase in dopamine (DA) efflux, inhibition of DA reuptake, or DA degradation by the enzyme monoamine oxidase (MAO).	Open-field test.	Frey et al. (2006).
Ouabain	Sodium potassium-activated adenosine triphosphatase (Na ⁺ , K ⁺ -ATPase) inhibitor.	Open-field test.	El-Mallakh et al. (1995).
Ketamine	Non-competitive NMDA glutamate receptor antagonist.	Open-field test.	Ghedim et al. (2012).
Environmental Sleep deprivation Genetic	Circadian rhythm disturbances.	Open-field test and aggressive behavior.	Benedetti et al. (2008).
Dopamine transporter (DAT)	Inactivation of the DAT causing alterations in DA transmission.	Open-field test.	Giros et al. (1966).
Circadian locomotor output cycle kaput (CLOCK)	Abnormalities in the circadian system.	Intracranial self-stimulation, sucrose preference, Forced swim test, Open-field test, Elevated plus maze test, Reward/Aversion Test, Learned helplessness following inescapable shock.	McClung et al. (2005). Roybal et al. (2007).
Glutamate receptor 6 (GluR6)	GluR6 gene knockout.	Passive avoidance test, Open-field test, Home cage activity, Social interaction, Elevated plus-maze, Forced swim test, Resident-intruder.	Shaltiel et al. (2008).
Extracellular signal-regulated kinase 1 (ERK1)	Disruption of the ERK1 gene.	Elevated plus-maze test, Forced swim test, Exploratory behavior, Long-term home-cage voluntary wheel running activity, Behavioral response to amphetamine, Monitoring the behavioral changes drug-induced.	Engel et al. (2009).
Postsynaptic density protein SHANK3	Overexpression of SHANK3.	Open-field test, Home-cage activity, Tail suspension test, Acoustic startle response and prepulse inhibition, Circadian rhythms, Three chamber test, Grooming, Ultrasonic vocalization, EEG measurement.	Han et al. (2013).
Neuron-specific α -3 subunit (ATP1A3)	Mutation in the neuron-specific NKA α 3 isoform.	Open-field test, Forced swim test, Elevated plus-maze test, Light-Dark Box, Head Tracking, Object and Holeboard Exploration.	Kirshenbaum et al. (2011).
B-cell lymphoma 2 (Bcl-2)	Bcl-2 heterozygote knockout.	Sucrose Preference Test, Acoustic Habituation, Startle, and PPI, EEG and Electromyographic Recordings of Sleep, Circadian Rhythms, Electrocorticography.	Lien et al. (2008).

Effects of natural compounds and derivatives in experimental models of mania

Experimental preclinical studies have revealed a range of complex psychotropic activity in natural compounds that are potentially beneficial for treating certain psychiatric conditions, such as manic episodes. Because oxidative stress has been associated with the etiology of BD, many experimental studies have focused on the use of antioxidants in the prevention of experimental manic episodes (Fig. 3).

Berry fruit extracts

Blueberry and blackberry fruit extracts, which are rich in polyphenols, have been investigated as therapeutic tools in animal models of mania. In a rat model of manic behavior induced by ketamine, Debom et al. (2016) and Spohr et al. (2019) reported that blueberry extract (*Vaccinium virgatum*, Ericaceae family) (200 mg/kg, administered for 14 days by gavage) prevented the manic-like behavior and attenuated oxidative damage in cerebral cortex, hippocampus, and striatum similarly to lithium treatment. In addition, blueberry extract was able to prevent the increase in Na^+ , K^+ -ATPase, and acetylcholinesterase (AChE) activity induced by ketamine in these brain regions (Spohr et al. 2019) (Table 2). Another similar study, using the mania model induced by ketamine, also showed that blackberry extract (*Rubus* sp., Rosaceae family) (200 mg/kg) prevented hyperlocomotion and oxidative brain damage in the cerebral cortex, hippocampus, and striatum (Chaves et al. 2020) (Table 2). In addition, both blueberry and blackberry extracts also modulated the serum and brain IL-6 levels in the ketamine mania model (Debom et al. 2016; Chaves et al. 2020). These

beneficial effects of these fruit extracts can be attributed to the presence of a large quantity of anthocyanins, a subset of flavonoids, which are important compounds with antioxidant and anti-inflammatory actions. Anthocyanins are the most abundant polyphenols in many red, blue, and purple fruits (Smeriglio et al. 2016), and appropriate intake of these phytochemicals is thought to promote health conditions and general well-being (Smeriglio et al. 2016; Fraga et al. 2019), as demonstrated in some studies examining brain health in humans (Boespflug et al. 2018). In fact, these red fruit extracts have been shown to modulate pathways involved in BD, and consumption of these extracts could be a useful alternative to prevent or reduce manic episodes, demonstrating its neuroprotective role in the central nervous system as a natural therapeutic approach (Debom et al. 2016, Spohr et al. 2019, Chaves et al. 2020).

Cecropia pachystachya extract

Cecropia pachystachya (Urticaceae family) is a plant that grows in the rainforests of South America and is popularly known in Brazil as embauba. It has been documented in the literature that *Cecropia pachystachya* presents many beneficial such as hypoglycemic (Aragão et al. 2010), antidepressant (Gazal et al. 2014a, b), anti-inflammatory, and antioxidant (Pacheco et al. 2014) actions. Gazal et al. (2015) demonstrated that *Cecropia pachystachya* aqueous extract (200 and 400 mg/kg) administered for 14 days prevented hyperlocomotion and oxidative brain damage in the hippocampus and prefrontal cortex in a mania model induced by ketamine (Table 2). These beneficial effects have been attributed to phytochemical compounds present in the

Fig. 3 Neuroprotective effects of natural compounds and derivatives in neurochemical changes found in mania experimental models

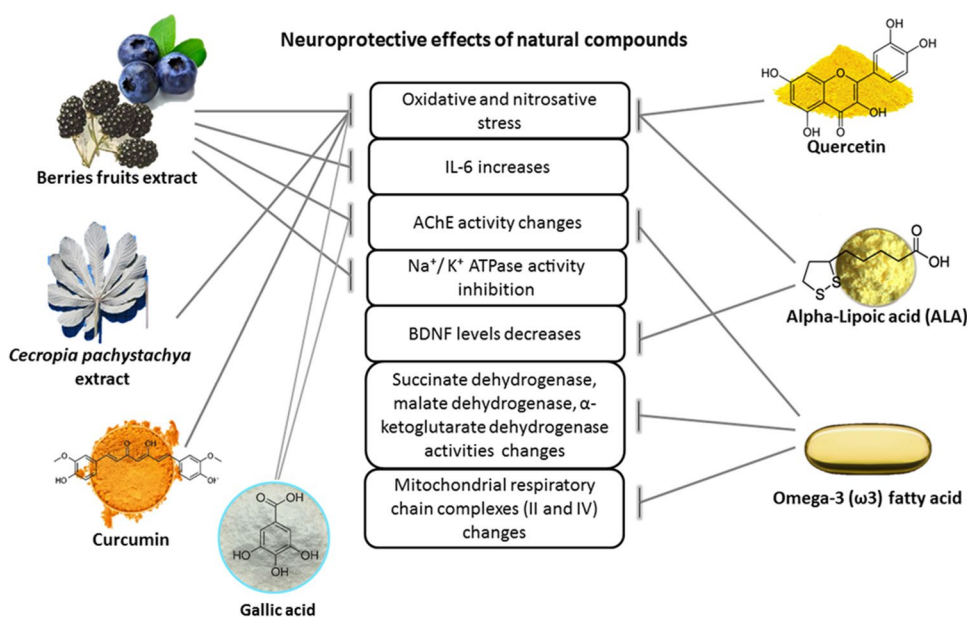


Table 2 Extracts from natural compounds investigated in preclinical and clinical models of mania

Extract	Source	Reference	Characterization	Protocol
Preclinical studies				
Blueberry extract (<i>Vaccinium virgatum</i>)	Fruits	Debom et al. (2016) Spohr et al. (2019)	Anthocyanin identification	200 mg/kg administered for 14 days by gavage
			Cianidin-3- <i>O</i> -hexoside	
			Cianidin-3- <i>O</i> -pentoside	
			Peonidin-3- <i>O</i> -hexoside	
			Delphinidin-3- <i>O</i> -hexoside	
			Delphinidin-3- <i>O</i> -pentoside	
			Petunidin-3- <i>O</i> -hexoside	
			Malvidin-3- <i>O</i> -hexoside	
			Anthocyanin identification	
			Delphinidin- <i>O</i> -galactoside	
			Delphinidin- <i>O</i> -glucoside	
			Cyanidin- <i>O</i> -galactoside	
			Delphinidin- <i>O</i> -pentoside	
			Cyanidin- <i>O</i> -glucoside	
			Petunidin- <i>O</i> -galactoside	
			Petunidin- <i>O</i> -glucoside	
			Cyanidin- <i>O</i> -pentoside	
			Peonidin- <i>O</i> -galactoside	
			Petunidin- <i>O</i> -pentoside	
			Peonidin- <i>O</i> -glucoside	
			Malvidin- <i>O</i> -galactoside	
			Peonidin- <i>O</i> -pentoside	
			Malvidin- <i>O</i> -pentoside	
Blackberry extract (<i>Rubus</i> sp.)	Fruits	Chaves et al. (2020)	Anthocyanin identification	200 mg/kg administered for 14 days by gavage
			Cyanidin- <i>O</i> -galactoside	
			Cyanidin- <i>O</i> -glucoside	
<i>Cecropia pachystachya</i> extract	Leaves	Gazal et al. (2015)	Cyanidin- <i>O</i> -rutoside	200 and 400 mg/kg administered for 14 days by gavage
			Pelargonidin- <i>O</i> -glucoside	
			Cyanidin- <i>O</i> -malonylhexoside	
			Chlorogenic acid	
			Isoorientin	
			Orientin	
			Isovitexin	
			Isoquercitrin	

Table 3 Isolated compounds investigated in preclinical and clinical models of mania

Compound	Source	Reference	Protocol
Preclinical studies			
Curcumin	<i>Curcuma longa</i> Family: Zingiberaceae	Gazal et al. (2014a)	20 and 50 mg/kg administered for 14 days by gavage
Quercetin	<i>Allium cepa</i> L. Family: Amaryllidaceae <i>Asparagus officinalis</i> L. Family: Asparagaceae <i>Lactuca sativa</i> L. Family: Asteraceae	Kanazawa et al. (2016, 2017)	10 and 40 mg/kg administered for 2 days by intraperitoneal route 2.5, 5, 10 and 40 mg/kg administered for 21 days by intraperitoneal route
Alpha-lipoic acid (ALA)	Muscle meats, heart and liver	Macêdo et al. (2012)	50 and 100 mg/kg administered for 7 days by gavage
(R)-(—)-carvone (S)-(+)-carvone	<i>Mentha spicata</i> Family: Lamiaceae <i>Carum carvi</i> Family: Apiaceae	Nogoceke et al. (2016)	50 and 100 mg/kg administered once (acute) or for 21 days (chronic) by intraperitoneal route
Omega-3 (ω 3) fatty acid	Oily fish and marine mammals	Arunagiri & Balamurugan (2016) Cancelier et al. (2017)	1.5 ml/kg administered for 7 days by gavage 0.8 g/kg administered for 14 days by gavage
Gallic acid	Red wine Green tea <i>Punica granatum</i> Family: Lythraceae	Recart et al. (2021)	50 and 100 mg/kg administered for 14 days by gavage

extract used, such as chlorogenic acid, isoorientin, orientin, isovitexin, and isoquercetin (Gazal et al. 2015).

Curcumin

Isolated natural compounds have also been investigated as therapeutic alternatives in different animal models of mania. Turmeric (*Curcuma longa*) is a plant related to the ginger family (Zingiberaceae) originated from India (Kotha and Luthria 2019). Curcumin (1,7-bis-(4-hydroxy-3-methoxyphenyl)-hepta-1,6-diene-3,5-dione) is one of bioactives curcuminoids of this plant (Kotha and Luthria 2019). Gazal et al. (2014a) demonstrated that pretreatment with curcumin (20 and 50 mg/kg) by the oral route for 14 days prevented the manic-like behavior caused by ketamine (Gazal et al. 2014a) (Table 3). This phytochemical decreased lipid peroxidation and increased thiol levels and the activity of antioxidant enzymes such as superoxide dismutase and catalase in the brain. The authors concluded that curcumin prevented oxidative damage, demonstrating its therapeutic potential in prophylactic interventions during the manic phase (Gazal et al. 2014a). Notably, curcumin regulates multiple targets that include transcription factors activated by environmental stressors and pro-inflammatory cytokines, protein kinases, enzymes, growth factors, inflammatory mediators, and anti-apoptotic proteins (Brietzke et al. 2013). Moreover, this compound has been studied in different disorders in preclinical models, such as Alzheimer's disease and Parkinson's disease, and

has demonstrated significant neuroprotective potential (Bassani et al. 2017; Sharma and Nehru 2018).

Quercetin

Quercetin (3,3',4',5,7-pentahydroxyflavone) is a common flavonol present in many foods (Costa et al. 2016). Among vegetables, the highest levels of quercetin have been found in onions (*Allium cepa* L.), asparagus (*Asparagus officinalis* L.), and red leaf lettuce (*Lactuca sativa* L.). In fruits, apples have the most expressive content, together with cherries and various berries (Costa et al. 2016). The antimanic-like effect of quercetin has been demonstrated in studies by Kanazawa and collaborators (2016, 2017). Quercetin, a flavonoid that possesses antioxidant properties and inhibits protein kinase C (PKC), prevented manic-like behavior induced by 24 h paradoxical sleep deprivation (PSD) (10 or 40 mg/kg, i.p. during an interval of 24 h) and by methylphenidate (5 mg/kg, i.p.)-induced hyperlocomotion (2.5, 5, 10, and 40 mg/kg, i.p. for 21 days) (Kanazawa et al. 2016, 2017) (Table 3). In both protocols, quercetin was able to protect against oxidative damage in several brain regions. These behavioral and neurochemical effects of quercetin appeared to be correlated with its ability to reduce oxidative stress. In addition, no tolerance was affected after repeated treatment, which is an important issue for the treatment of mania (Kanazawa et al. 2016, 2017). Since an increase in PKC and oxidative stress has been related to mania, compounds with antioxidant effects and/or inhibitory actions on PKC

are thought to be important for preventing manic effects. Quercetin exhibits these two properties, which makes it an important target for the study of this pathology (Kanazawa et al. 2016, 2017).

Alpha-Lipoic acid (ALA)

Alpha-lipoic acid (ALA) is a natural compound synthesized in plants and animals in the mitochondria from octanoic acid, and it has been identified as a catalytic agent for oxidative decarboxylation of pyruvate and α -ketoglutarate. ALA is also absorbed intact from dietary sources and transiently accumulates in many tissues. Dietary sources include muscle meats, heart and liver, and to a lesser degree, fruits and vegetables (Shay et al. 2009). It has been described as a potent antioxidant compound and a modulator of inflammatory pathways. Macêdo et al. (2012) used an amphetamine-induced (AMPH) model of mania in mice to investigate the preventive and reversal effects of oral administration of ALA (50 or 100 mg/kg) (Table 3). ALA prevented and reversed AMPH-induced hyperlocomotion and oxidative damage in the brain in both protocols. Moreover, ALA in the reversion model increased BDNF levels in the hippocampus. These findings showed that ALA was effective in reversing and preventing AMPH-induced behavioral and neurochemical damage, possibly by scavenging free radical and lipid peroxidation products. Therefore, this compound demonstrated antimanic-like effects and could be considered an adjunct treatment for BD (Macêdo et al. 2012).

Monoterpenes

Nogoceke et al. (2016) evaluated (R)-(-)-carvone and (S)-(+)-carvone (5-isopropenyl-2-methyl-2-cyclohexenone), two types of monoterpene abundants in many plants, including in spearmint (*Mentha spicata*) and caraway (*Carum carvi*), respectively (Asle-Rousta et al. 2020). Two protocols of mania were executed in mice: acute and chronic (21 days) carvone (50 and 100 mg/kg, i.p.) treatment in a model induced by methylphenidate (5 mg/kg, i.p.), and acute carvone treatment during sleep deprivation. Acute and chronic treatment with (S)-(+)-carvone (S-100 mg/kg) and (R)-(-)-carvone (R-100 mg/kg) in the methylphenidate model was able to decrease mouse locomotion. S-100 and R-100 also blocked sleep deprivation-induced hyperactivity (Table 3). The antimanic-like action of both enantiomers of carvone in two different protocols minimized the probability of false-positive results, which was a significant behavioral outcome. The mechanism of action of carvone might be related to the blockade of voltage-gated sodium channels and/or GABAergic activity, which are related to other

biological effects of this compound, such as anticonvulsant and antinociceptive actions (Nogoceke et al. 2016).

Omega-3 (ω 3) fatty acid

Omega-3 polyunsaturated fatty acids (PUFAs) are essential fatty acids, which include α -linolenic acid (ALA; 18:3 ω -3), stearidonic acid (SDA; 18:4 ω -3), eicosapentaenoic acid (EPA; 20:5 ω -3), docosapentaenoic acid (DPA; 22:5 ω -3), and docosahexaenoic acid (DHA; 22:6 ω -3) (Shahidi and Ambigaipalan 2018). These compounds are exclusively found in aquatic organisms, such as oily fish and marine mammals (Shahidi and Ambigaipalan 2018). The effect of ω 3 fatty acids has also been investigated in a model of mania induced by methylphenidate (MPD 5 mg/kg, i.p.) administered for 14 days. Omega-3, lithium and aripiprazole were administered from 8th day onwards. Treatment with ω 3 (1.5 ml/kg, oral route) and ω 3 + lithium + aripiprazole protects against the MPD-induced decrease in succinate dehydrogenase, malate dehydrogenase, and α -ketoglutarate dehydrogenase activity in the brain (Table 3). Changes in the tricarboxylic acid (TCA) cycle is thought to alter brain metabolism and enhance free radical production, contributing to the physiopathology of manic-like behavior. With regard to AChE activity, an excess of acetylcholine can also disrupt brain mitochondrial activity and stimulate oxidative stress. Interestingly, the activity of these enzymes had a more promising effect in a group treated with ω 3 associated with standard drugs for BD. ω 3 supplementation can thus be considered in therapies, in combination with atypical antipsychotics and mood stabilizers, in order to optimize the clinical effectiveness of conventional treatment (Arunagiri and Balamurugan 2016; Arunagiri et al. 2014a, b).

Cancelier et al. (2017) also investigated ω 3 fatty acid treatment in animals subjected to mania model. Rats were treated with ω 3 (0.8 g/kg) by oral gavage, by itself or associated with either lithium (47.5 mg/kg i.p.) or valproate (VPA, 200 mg/kg i.p.) once a day for 14 days. Between days 8 and 15, animals received one daily injection of fenproporex (FEN, 12.5 mg/kg) or vehicle. In the reversal treatment, animals received FEN for 15 days, and between days 8 and 15 received the ω 3 by oral gavage, by itself or associated with either lithium or VPA once a day (Cancelier et al. 2017) (Table 3). ω 3 treatment co-administered with mood stabilizers prevented and reversed the effects of fenproporex on energy metabolism, as evaluated by succinate dehydrogenase (SHD) activity and mitochondrial respiratory chain complexes (II and IV) in the hippocampus, and by examining behavior parameters. This study reinforces the notion that this natural compound can exert protective effects on energy metabolism through its antioxidant capacity and may

be useful for the adjuvant treatment of BD (Cancelier et al. 2017).

Gallic acid

Gallic acid (3,4,5-trihydroxybenzoic acid) is one of the most abundant phenolic acids in the plant kingdom, with various pharmacological properties already described in literature (Kahkeshani et al. 2019). This compound is found in red wine, green tea, and is distributed across the plant kingdom, being present in leaves (beriberri), roots and bark (pomegranates and gall nuts) (Choubey et al. 2015). Recart et al. (2021) investigated gallic acid (50 and 100 mg/kg) or lithium (45 mg/kg twice a day) treatment administered for 14 days in a rat model of ketamine-induced mania (Table 3). Similar to lithium, gallic acid intake decreased the hyperlocomotion in behavioral assessment. This phytochemical also ameliorates brain oxidative damage and AChE activity. Thus, gallic acid demonstrated relevant activities in manic-like behavior via reduction of oxidative stress and prevention of cholinergic signaling dysfunction in the brain.

Here, we showed that natural extracts as well as isolated bioactive compounds have relevant therapeutic actions in experimental models of manic-like behavior (Fig. 3). Berries fruits and *Cecropia* extracts, curcumin, quercetin, gallic acid, omega-3 (ω 3) and ALA are capable to interact with multiple brain targets that are related to the pathogenesis of mania behavior such as, oxidative stress, ion pumps activities, cholinergic signaling, neuroinflammation, neurotrophic factors and energetic metabolism (Fig. 3). As regards plant extracts such as blueberry, blackberry and *Cecropia* it is important to consider that the biological activity may result

from the combination of the several active compounds. In fact, sometimes a mixture of compounds can have a better effect than an equivalent dose of the single isolated substance. In this line, synergistic mechanisms between the compounds may be responsible to enhance, amplify or potentiate the therapeutics effects (Liu 2004).

In addition, in many of these preclinical studies, standard drugs used for BD treatment also were used for comparison and validation of the experimental mania model. Lithium was used as a positive control, alone or in combination with the natural compounds. It is important to discuss that many of the natural compounds and derivatives evaluated have demonstrated a very similar effect to the standard treatment. It should be noted that studies also have demonstrated the antioxidant activity of natural compounds and lithium (Debom et al. 2016; Chaves et al. 2020; Gazal et al. 2014a, b, 2015; Kanazawa et al. 2016; Macêdo et al. 2012; Nogoceke et al. 2016). In fact, the antioxidant role of lithium has also been described by others studies from the literature (Frey et al. 2006; Khairova et al. 2012).

Three studies also evaluated the association of natural compounds with standard drugs. In this context, Spohr et al. (2019) demonstrated that the preventive effects were decreased with the association of blueberry extract and lithium in a model of mania. A possible explanation is the interaction of conventional drugs with several bioactive compounds found in blueberry extract (Spohr et al. 2019). In contrast, the association of omega-3 (ω 3) fatty acid with conventional drugs has more pronounced effects against changes induced by animal model of mania, suggesting that ω 3 might be an important adjunctive therapy in BD (Arunagiri and Balamurugan 2016; Cancelier et al. 2017).

Table 4 Isolated compounds evaluated in clinical studies with BD patients

Clinical studies		
Compound	Source	Reference
Omega-3 (ω 3) fatty acid	Oily fish and marine mammals	Vesco et al. (2015) Clayton et al. (2009) Frangou et al. (2006) Osher et al. (2005)
Inositol	<i>Prunus dulcis</i> Family: Rosaceae <i>Juglans</i> sp. Family: Juglandaceae <i>Bertholletia excelsa</i> Family: Lecythidaceae	Eden Evins et al. (2006) Chengappa et al. (2000)
N-acetylcysteine (NAC)	<i>Allium cepa</i> L. Family: Amaryllidaceae L-cysteine Endogenous proteins breakdown	Dean et al. (2012) Magalhães et al. (2011a) Magalhães et al. (2011b) Magalhães et al. (2013) Berk et al. (2011a) Berk et al. (2008)
Vitamin D	Cholecalciferol (vitamin D3) - animal sources Ergocalciferol (vitamin D2) - plant sources	Sikoglu et al. (2015)

In this sense, is plausible consider that experimental models of mania become important tools to better understand the pathways involved in the effects elicited by natural products and derivatives and thus enable the transposition to humans with BD.

Evaluation of natural compounds and derivatives in clinical BD studies

Accumulating evidence from several decades of research and utilization of several bioactive compounds from both plant and non-plant sources have led to the discovery of potential therapeutic agents. This has created opportunities for advancements in the prevention and treatment of psychiatric diseases. The effects of natural compounds and derivatives have been widely investigated in patients with psychiatric disorders such as major depression; however, little data exist about the beneficial effects of these compounds in patients with BD (Table 4). We will present an overview of advances made in this field over the last several decades, as well as the role of natural compounds and derivatives and their therapeutic potential in clinical studies involving individuals with BD.

Omega-3 (ω 3) fatty acid

ω 3 is a long-chain fatty acid (14 to 22 carbon atoms), the polyunsaturated type and contains the first double bond in carbon 3, from the methyl radical. ω 3 is considered a functional food, which can be found in both natural forms (marine animals from cold regions, mainly salmon, tuna and trout) and artificial forms (drugs) (Layé et al. 2018; Healy-Stoffel and Levant 2018) (Table 4).

The benefits of omega-3 (ω 3) fatty acids for mood disorders have been reported in the literature. ω 3 fatty acids, especially those present in neuronal membranes, are essential for the proper functioning of the central nervous system. Moreover, ω 3 deficiency has been associated with neuroinflammation, a condition present in BD. Accordingly, previous research on eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) has suggested that polyunsaturated fatty acids (PUFAs) may improve brain function in patients with BD (Vesco et al. 2015).

The effectiveness of ω 3 fatty acids against manic episodes and other symptoms involved in BD have previously been evaluated in clinical studies. Vesco et al. (2015) demonstrated in a case involving ω 3 supplementation (1000 mg in the morning and 1500 mg at night) that this fatty acid can be a safe adjuvant intervention for the treatment of BD in children and adolescents. A 2-year follow-up, beginning from the age of 12, offers hope of a lasting benefit, since a

significant and sustained clinical improvement in depression and psychological symptoms was observed (Vesco et al. 2015). In addition, Osher et al. (2005) demonstrated that 8 of the 10 patients who completed at least 1 month of 1.5 to 2 g/day of ω 3 supplementation achieved a 50% or greater reduction in Hamilton Rating Scale for Depression (HRSD) scores within 1 month (Osher et al. 2005). Similar results were demonstrated by Frangou et al. (2006), who showed that in a 12-week double-blind study involving individuals with bipolar depression, adjunctive treatment with 1 g/day or 2 g/day of ethyl-eicosapentaenoic acid (ethyl-EPA) significantly improved clinical symptoms compared with that of a placebo (Frangou et al. 2006).

Moreover, in children and adolescents with juvenile BD supplemented with 360 mg/day of EPA and 1560 mg/day of DHA for 6 weeks in an open-label study, the clinician ratings of mania and depression were significantly lower, and global functioning was significantly higher, after supplementation. In fact, a reduction in mania and depression in juvenile BD was associated with long-chain omega-3 polyunsaturated fatty acid supplementation (Clayton et al. 2009).

It should be noted that several studies in the literature have demonstrated the therapeutic potential of ω 3 in neuropsychiatric diseases, especially due to its biological effects on the brain. For example, DHA constitutes approximately 12–15% by weight of the total fatty acids in the human brain (Layé et al. 2018; Healy-Stoffel and Levant 2018). Thus, this fatty acid might contribute to the maintenance of homeostasis in conditions where there is a disruption in the physicochemical properties of neuronal membranes, as has been reported in BD.

In addition, ω 3 is known to be a potent anti-inflammatory agent. Considering that the inflammatory process participates in the pathophysiology of BD, the presence of compounds capable of minimizing neuroinflammation is crucial. A mechanism by which ω 3 plays an anti-inflammatory role is through the downregulation of inflammatory gene expression, such as those of cytokines or enzymes involved in the synthesis of eicosanoids. In humans, ω 3 consumption is associated with a lower risk of inflammation-associated neurologic disorders (Layé et al. 2018; Healy-Stoffel and Levant 2018). For this reason, new studies aimed at investigating the neuroprotective effects of ω 3 in patients with BD are extremely important and should be encouraged in order to improve the quality of life of patients with this pathology.

Inositol

Inositol is a natural compound found in several foods (fruits, dark green vegetables, eggs, nuts and different types of seeds) and is present in all animal tissues, mainly in the heart and brain. This compound constitutes the intracellular phosphatidylinositol, a second messenger system

associated with neurotransmitter receptors that include serotonin, dopamine, and glutamate receptors. It is known that soluble inositol phosphate controls the release of Ca^{2+} from intracellular stores. Accordingly, inositol has been reported to be effective for treating patients with depression and has been tested as an alternative for BD patients (Formoso et al. 2019; Schneider 2015; Harwood 2005).

Chengappa et al. (2000) investigated the potential efficacy and safety of inositol in BD. Men and women with bipolar depression were randomly assigned to receive either 12 g of inositol or D-glucose as a placebo for 6 weeks. The authors found that 50% of patients treated with inositol responded with a 50% or higher reduction in the initial score of the Hamilton Depression Rating Scale and the Clinical Global Improvement scale (Chengappa et al. 2000). In addition, another study evaluated the efficacy of adjunctive inositol (5–20 g/day) for bipolar depression in a placebo-controlled trial study. Seventeen participants with bipolar depression were randomized to receive double-blind inositol or placebo for 6 weeks. The response to inositol was highly variable. Of the nine individuals that received inositol, two showed a worsening of > 50% in HRSD scores without the final treatment, three exhibited no changes, and four had an improvement > 50%, but there was no significant effect between groups. Based on these results, the authors suggested further studies to determine whether inositol can be effective in a subpopulation of patients that do not exhibit agitation or aggressive behavior and whether inositol may be more effective in treatment-resistant patients (Evins et al. 2006).

An important aspect to be taken into account is the reduction in inositol concentrations in the rat brain induced by mood stabilizers, such as lithium and valproate. Therefore, the use of inositol as an adjuvant in the treatment of BD would be an interesting approach for the management of psychiatric patients. Furthermore, in recent years, antioxidant and anti-inflammatory effects exerted by this compound have been reported, which has increased interest in research studies of inositol in BD, since oxidative stress and inflammation are involved in the pathophysiology of this disorder (Formoso et al. 2019; Schneider 2015; Harwood 2005).

N-acetylcysteine (NAC)

N-acetylcysteine (NAC) is an n-acetylated derivative of cysteine (non-essential amino acid found mainly in milk and dairy products, meat, eggs and whole grains) that contains a sulfhydryl group. NAC is able to bind to metals and scavenger free radicals, in addition to being able to directly reduce H_2O_2 and $\text{O}_2^{\bullet -}$ to less reactive species forming sulfuric radicals or cysteine (Deepmala et al. 2015; Samuni et al. 2013) (Table 4). NAC is a thiol precursor of cysteine and reduced glutathione, so considering the antioxidant effects of NAC, many studies have investigated its benefits

involving multiple mechanisms of action across neurological disorders, including BD (Deepmala et al. 2015; Samuni et al. 2013).

Berk et al. (2011a, b) conducted a study with 132 participants with BD completing an 8-week open label treatment phase with administration of 1 g of NAC twice daily. Treatment with NAC reduced the Bipolar Depression Rating Scale (BDRS) score. However, a placebo arm, which is essential to make definitive conclusions regarding efficacy, was absent in this investigation (Berk et al. 2011a, b). In another randomized, double-blind, multicenter, placebo-controlled study, 75 individuals with BD were treated with 1 g of NAC twice daily adjunctive to usual medication over 24 weeks. NAC treatment significantly improved clinical symptoms (Berk et al. 2008). There was no effect of NAC on time-to-mood episodes and no significant between-group differences in adverse events. Moderated functional outcomes, but not depression, were found in this protocol.

In 2012, Dean and colleagues conducted a randomized, double-blind, placebo-controlled trial study, and they reported that treatment with NAC compared with placebo did not prevent changes in cognitive function tested at baseline and after 6 months to assess changes in cognitive function following administration of either 2000 mg of NAC daily or a placebo. However, this study was conducted with a small sample size and a limited battery of cognitive tests, which limits the data obtained (Dean et al. 2012).

Some researchers developed a series of studies whose objective was to investigate the effects of NAC (500 mg) administration twice daily for 24 weeks in patients with BD. The main findings of the studies were as follows: (1) Six patients in the group treated with NAC ($n = 7$) showed total remission of depressive and manic symptoms. On the other hand, two patients in the placebo group also showed complete remission of depressive symptoms ($n = 7$ in the group placebo) (Magalhães et al. 2011a). (2) Eight of the ten participants who received NAC showed alleviation of clinical symptoms of depression, while only one of the seven participants allocated to the placebo showed the same result (Magalhães et al. 2011b). (3) The analysis of this study pointed to an improvement in the group that received NAC for manic symptoms and a worsening of depressive symptoms in the placebo group (Magalhães et al. 2013).

The main mechanisms underlying NAC function that have been investigated are related to glutathione, such as antioxidant properties and modulation of redox-regulated signal transduction associated mainly with its role in the storage and transport of cysteine, regulation of cell proliferation, immune responses, leukotriene and prostaglandin metabolism, detoxification of electrophilic xenobiotics, and synthesis of deoxyribonucleotides. In addition, several applications of NAC have been proposed especially considering its ability to minimize oxidative stress, infections,

Table 5 Extracts from natural compounds evaluated in clinical studies with BD patients

Clinical studies			
Extract	Source	Reference	Characterization
Free and Easy Wanderer Plus (FEWP)	Dry root, fresh root, dry ripe fruit, dry root bark, stem, dry sclerotium, dry stem, leaf	Zhang et al. (2007b)	<i>Bupleurum chinense</i> DC. <i>Scutellaria baicalensis</i> Georgi <i>Zingiber officinale</i> Rosc. <i>Angelica sinensis</i> (Oliv.) Diels. <i>Gardenia jasminoides</i> Ellis. <i>Paeonia suffruticosa</i> Andr. <i>Paeonia lactiflora</i> Pall. <i>Atractylodes macrocephala</i> Koidz. <i>Poriae Cocos</i> (Schw.) Wolf <i>Mentha haplocalyx</i> Briq. <i>Glycyrrhiza uralensis</i> Fisch.
<i>Withania somnifera</i> standardized extract	Leaves and roots	Chengappa et al. (2013)	Withanolides Oligosaccharides Withaferin A

and inflammatory conditions (Deepmala et al. 2015; Slattery et al. 2015; Samuni et al. 2013). In view of all the effects mediated by NAC in addition to its clinical effects already demonstrated, this natural compound presents itself as a relevant therapeutic alternative for BD.

Vitamin D

1,25 (OH)₂D₃ (vitamin D) is produced endogenously and depend on dietary intake of calcium and phosphorus, age, sex, skin pigmentation and sun exposure. Among the main dietary sources of this vitamin are fish, butter, beef liver and chicken, eggs and milk (Altunsoy et al. 2018) (Table 4). Vitamin D has been recognized as a neurosteroid that modulates multiple brain functions and plays a pivotal role in brain development, neurotransmission, neuroprotection, and immunomodulation (Cui et al. 2017).

Vitamin D₃ supplementation also appears to be a promising natural agent for the treatment of mania. BD patients supplemented with 2000 IU of vitamin D₃ exhibited an improvement in mood symptoms in conjunction with their brain neurochemistry (Sikoglu et al. 2015). In recent years, several studies on the neuroprotective effects of vitamin D have been carried out, including as a strategy for healthy brain aging, since augmenting brain BDNF appears to be a key mechanism through which this vitamin counteracts age-related brain dysfunction (Beckmann et al. 2019).

Changes in serum vitamin D levels in patients with BD have been previously reported in the literature. Altunsoy et al. (2018) showed an important correlation between vitamin D deficiency and acute manic episodes. Unfortunately, this interaction over long periods has not been demonstrated. Therefore, taking into account the data indicating that BD

patients may be deficient in vitamin D, along with the promising and beneficial effects of this vitamin, improved studies that aim to investigate the potential of this compound in BD should be encouraged.

Free and easy wanderer plus

Free and Easy Wanderer Plus (FEWP) is a polyherbal preparation which therapeutic benefits have been extensively evaluated in patients with various depressive disorders. However, the effects of FEWP on BD patients are still poorly investigated (Qin et al., 2011). So, Zhang et al. (2007a) demonstrated, through a systematic investigation, a number of beneficial effects of the herbal preparation FEWP, a Chinese herbal medicine (36 g/day for 12 weeks) used as an adjunct to carbamazepine (CBZ) in the treatment of bipolar patients, compared to CBZ monotherapy and placebo. They suggested that adjunctive FEWP has additive beneficial effects in bipolar patients, particularly for those in the depressive phase (Zhang et al. 2007a). The same authors suggested that FEWP monotherapy may also be an effective alternative treatment for depressed conditions after 26 weeks of treatment with FEWP under double-blind conditions. In this study, they demonstrated that in addition to a significantly lower overall follow-up rate at the end point compared to placebo, patients who received FEWP in addition to CBZ showed significantly fewer adverse effects and lower serum CBZ levels than those on placebo. In addition, FEWP (36 g/day for 12 weeks) was used as monotherapy in patients with depression (Zhang et al. 2007b) (Table 5). The results suggest that adjuvant FEWP improves CBZ tolerability upon long-term use, in addition to its use as an effective alternative form of monotherapy for depressive conditions (Zhang et al. 2007b).

Withania somnifera

Withania Somnifera (WSE), belonging to the family Solanaceae, is an Ayurvedic herb also known as Indian winter cherry and Indian ginseng that has been traditionally known in India for its numerous beneficial effects on health. WSE is one of the most important herbs in Ayurveda, which has been used for > 3000 years in stress management, energy elevation and improving cognitive health and to lower inflammation, blood sugar levels, cortisol, anxiety, and depression (Dutta et al. 2019). Cognitive impairment in patients with BD contributes to inadequate functional recovery. However, despite this knowledge, the mechanisms behind cognitive reduction are still poorly understood, and sometimes end up being neglected during pharmacological treatment (Chengappa et al. 2013). For this reason, Chengappa et al. (2013) investigated the effects of *Withania somnifera* (WSE) extract, which is known to increase body resistance to stress and diseases (Table 5).

Sixty euthymic subjects with DSM-IV bipolar disorder were enrolled in an 8-week, double-blind, placebo-controlled, randomized study of WSE (500 mg/day) as a procognitive agent added adjunctively to medications being used as maintenance treatment for BD. The authors observed that treatment with WSE provided significant benefits for three cognitive tasks: digit span backward, flanker neutral response time, and the social cognition response rating of the Penn Emotional Acuity Test compared with the control group (Chengappa et al. 2013).

Thus, the authors reported that WSE would be an interesting alternative for managing cognitive changes in patients with BD. Corroborating with the beneficial effects observed in this clinical study, preclinical studies with rodents have reported that WSE is able to protect against memory deficits in addition to inducing regeneration of dendritic spines and axons in mice, and that WSE attenuates damage to hippocampal neurons in the CA2 and CA3, decreases hypercortisolemia, and shows

Table 6 Effects of natural compounds and derivatives in BD patients

Compound	Reference	Protocol	Effects
Omega-3 (ω 3) fatty acid	Vesco et al. (2015) Clayton et al. (2009) Frangou et al. (2006) Osher et al. (2005)	1000 mg in the morning and 1500 mg at night for 2 years	Improves clinical depression and psychological symptoms Improves clinical ratings of mania and depression Improves clinical symptoms compared with placebo
		360 mg/day of EPA and 1560 mg/day of DHA for 6 weeks 1 g/day or 2 g/day for 12 week 1.5 to 2 g/day for month	Improves the Hamilton Rating Scale for Depression (HRSD) score
Inositol	Eden Evins et al. (2006) Chengappa et al. (2000)	5–20 g/day for 6 weeks	Improves the Hamilton Rating Scale for Depression (HRSD) scores; no significant effect was found between treated and placebo groups.
		12 g/day for 6 weeks	Reduces the initial score of the Hamilton Depression Rating Scale and the Clinical Global Improvement scale; no significant effect was found between treated and placebo groups
N-acetylcysteine (NAC)	Dean et al. (2012) Magalhães et al. (2011a) Magalhães et al. (2011b) Magalhães et al. (2013) Berk et al. (2011a) Berk et al. (2008)	2000 mg daily for 6 months	Did not prevent changes in cognitive function tested
		Two NAC (500 mg) capsules twice daily for 24 weeks	Remission of depressive and manic symptoms Improves the clinical symptoms of depression
		Two NAC (500 mg) capsules twice daily for 24 weeks	Improves manic symptoms
		Two NAC (500 mg) capsules twice daily for 24 weeks	Reduces the Bipolar Depression Rating Scale (BDRS) score
		Two NAC (500 mg) capsules twice daily for 24 weeks	Improves clinical symptoms; there was no effect of NAC on time-to-mood episode
		1 g twice daily for 8 weeks 1 g twice daily for 24 weeks	
Vitamin D	Sikoglu et al. (2015)	2000 IU/day for 8 weeks	Improves mood symptoms and brain neurochemistry changes
Free and Easy Wanderer Plus	Zhang et al. (2007b)	36 g/day for 12 weeks	Ameliorates the depressive phase
<i>Withania somnifera</i>	Chengappa et al. (2013)	500 mg/day for 8 weeks	Improves cognitive impairment

procholinergic but not glutamatergic or GABAergic effects in the rat brain (Chengappa et al. 2013). These beneficial effects are possibly associated with some bioactive components found in the WSE extract, such as steroidal lactones, termed glycowithanolides and sitoinosides, as well as Withaferin, which has antioxidant and neuroprotective properties (Chengappa et al. 2013).

Natural products and bioactive compounds are widely studied as a pharmacology alternative to prevention or treatment to several neurological disorders. Although the studies evaluating the effects of the natural compounds in BD patients are scarce, herein we showed that both isolated compounds such as omega 3, inositol, NAC and vitamin C and crude plants extracts were capable to improve the patients symptoms (measured by different scores) associated with BD (Table 6).

As regards clinical studies, there are some important points to be mentioned as the limited clinical psychological assessments and the variation among the patients themselves, such as age, sex, race, other pre-existing morbidities and the use of different medications. Nevertheless, here it is worth noting that as far as age is concerned, studies have been conducted in the two age groups most affected by mood changes in BD, adults and young people, which highlights the action of natural compounds in this disorder. Additionally, studies focusing on the mechanisms of action of natural compounds in BD clinical studies are still scarce.

On the other hand, it should be noted that no adverse effects were reported in the studies cited. Thus, new researches should continue to address the beneficial effects of natural compounds and their derivatives on BD, which can prove to be a challenge, mainly because there is a wide variety of compounds and consequently a difficulty of investigating their mechanisms of action. However, considering the effects demonstrated so far, this is a relevant research field, since it can be a therapeutic alternative for BD, improving the quality of life of patients.

Conclusions

BD is a complex medical condition that usually requires lifelong treatment. Preclinical and clinical studies have confirmed the potential of natural compounds and derivatives in both the prevention and management of BD phases. These compounds are beneficial in alleviating symptoms in patients as well as exerting neuroprotective effects in manic animal models through antioxidant, anti-inflammatory, anticholinesterase, and neurotrophic pathways. Due to the limited effectiveness and safety of

conventional drugs used in the treatment of BD, there is an urgent need for more effective, better-tolerated treatments without harmful side effects. Thus, natural compounds and derivatives have emerged as potential neuroprotective agents for BD; however, many aspects remain to be explored both at clinical and preclinical levels, especially during the manic phase. Finally, given the wide variety of natural products and derivatives and a limited number of studies with BD, there are numerous perspectives in this field for future studies.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval This article does not contain any studies with human participants by any of the authors. All studies performed by the authors with animals and cited in this manuscript has been previously published in accordance with the ethical standards of the institutional research guidelines for the care and use of animals.

Consent to participate Not applicable.

Conflicts of interest Authors declare no conflicts of interest.

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